

Taking to the skies

For centuries, people dreamed of taking to the air with birdlike wings, but attempts at winged flight usually ended in injury or death. Powered, controlled flight took a long time to engineer, and when it arrived, the pace of change was rapid. American brothers Orville and Wilbur Wright played a key role in this. The 1903 Wright Flyer—the first powered aircraft—flew 118 ft (36 m) on its maiden voyage in 1903, but just 11 years later, German aviator Karl Ingold flew a Mercedes Aviatik-Pfeil nonstop across 1,055 miles (1,699 km).

“To invent an airplane is nothing. To build one is something. But to fly is everything.”

Otto Lilienthal



Lilienthal with his glider in the mid-1890s

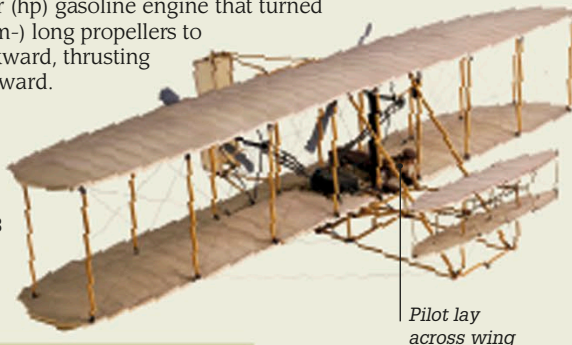
Gliding ahead

A scientific study of the principles of flight by English engineer George Cayley and German aviation pioneer Otto Lilienthal, among others, heralded the development of unpowered aircraft, which could glide short distances. Lilienthal made more than 2,000 glider flights, some going as far as 820 ft (250 m). His pioneering experiments took place between 1891 and his death in a gliding crash in 1896.

Powered flight

Inspired by Otto Lilienthal, Orville and Wilbur Wright, popularly known as the Wright Brothers, studied all aspects of gliders and flight before constructing the first powered aircraft, the Wright Flyer. It had a 12-horsepower (hp) gasoline engine that turned two 8-ft- (2.4-m-) long propellers to deflect air backward, thrusting the aircraft forward.

Model of 1903 Wright Flyer



Pilot lay across wing

Avro 504 biplanes being manufactured at a factory in Hampshire, England, during World War I



Key events

1903

The Wright Flyer made the first controlled flight by a powered aircraft at Kill Devil Hills, North Carolina. The pilot, Orville Wright, lay down on his stomach to reduce drag and the flight lasted 12 seconds.

1909

French aviator Louis Blériot was the first to fly the English Channel in his Blériot XI monoplane, which had a top speed of just 47 mph (75.6 km/h).

1911

The first take-off and landing on a ship were both performed by American aviator Eugene Ely piloting a Curtiss biplane. He took off from and landed on the USS *Pennsylvania* in San Francisco Bay, California.

1927

American pilot Charles Lindbergh made the first solo nonstop crossing of the Atlantic in his *Spirit of St. Louis* monoplane. Around 451 gallons (1,700 liters) of fuel were put into its wings and fuselage.

